

Multiple Choice questions

1. When setting up a local web development environment, which protocol is typically used by the browser to request web pages?

- a) FTP
- b) SSH
- c) HTTP
- d) SMTP

2. Which HTML5 element provides a container for navigation links?

- a) <menu>
- b) <nav>
- c) <section>
- d) <aside>

3. What is the semantic difference between and tags?

- a) No difference
- b) implies importance, is purely visual
- c) implies importance, is stylistic
- d) Both are deprecated

4. In Flexbox, which property allows a flex item to shrink if necessary?

- a) flex-grow
- b) flex-shrink
- c) flex-basis
- d) align-content

5. In CSS Grid, what does repeat (auto-fit, minmax(200px, 1fr)) achieve?

- a) Fixes grid columns at 200px
- b) Creates flexible columns that fill available space
- c) Adds spacing between grid items
- d) Aligns items to the left

6. What advantages do frameworks like Bootstrap or Tailwind CSS provide?

- a) Smaller file sizes
- b) Predefined responsive design and faster prototyping
- c) Built-in database connectivity
- d) Browser-based version control

7. When optimizing images for the web, which factor affects both quality and loading speed?

a) File path

c) Color depth

b) Resolution

d) File size

8. Which of the following selectors is the most expensive for the browser to render (least performant)?

a) menu li a

c) a:hover

b) #menu a

d) div > a

9. What will be logged?

```
const {a=5, b=10 }={a: 7};
```

```
console.log(a, b);
```

a) 5 10

c) 7 undefined

b) 7 10

d) undefined 10

10. What will this code print?

```
async function fetchData() {
```

```
  console.log('1');
```

```
  await Promise.resolve();
```

```
  console.log('2');
```

```
}
```

```
fetchData();
```

```
console.log('3');
```

a) 1-2-3

c) 3-1-2

b) 1-3-2

d) 2→1→3

11. What is the purpose of having an ICT code of conduct in a workplace environment?

Ans: Ensures responsible, ethical, and professional behavior in the workplace when using ICT systems. Promotes data protection, respect, and integrity.

12.Describe how to use mail merge in a word processing application and its typical use case.

Ans: Combines a template document with a data source (like names or addresses) to produce multiple personalized documents such as letters or certificates.

13.What steps are involved in setting up a basic folder structure for a web design project?

Ans: Create folders like images, css, js, and assets. Keep HTML files in the root directory to maintain organization and easy management.

14.What is the difference between the <section>, <article>, and <div> elements?

Ans:

- Section groups related content.
- Article is for independent, self-contained content (like a blog post).
- Div is a generic container mainly used for layout or styling.

15.Explain the difference between raster and vector images with one example each.

Ans: Raster images (JPEG, PNG) are made of pixels and lose quality when scaled.

Vector images (SVG, AI) use paths and can scale without losing quality.

16.What is the difference between relative units (like em, rem) and absolute units (like px) in CSS?

Ans: Relative units (em, rem) scale based on parent or root font size.

Absolute units (px) are fixed and do not depend on other elements.

17.Define an array and explain how looping through an array can be done using the forEach() method.

Ans: An array stores multiple values. The `forEach` method executes a function once for each array item.

Example:

```
const nums = [1,2,3];  
nums.forEach(n=> console.log(n));
```

18. What is the use of the `addEventListener()` method, and how does it differ from using the `onclick` attribute?

Ans: `addEventListener` allows multiple event handlers and offers better flexibility. `onclick` can only handle one function at a time for an event.

19. Explain the purpose of the `forEach()` and `map()` methods in JavaScript. How do they differ in output?

Ans: `forEach` executes a function on each item but does not return a value.

`map` executes a function and returns a new array.

20. Explain how the z-index property works and why it may not work as expected in some cases.

Ans: Controls element stacking order. May not work if element has static position or wrong stacking context.

21. What is the difference between `==` and `===` operators in JavaScript, and why is it generally better to use `===`?

Ans: compares values with type conversion.

compares both type and value.

It is safer to use for accurate comparisons.

22. Explain how Flexbox helps in creating responsive web layouts.

Ans: Flexbox makes layouts flexible and automatically adjusts element sizes and positions for different screen widths.

23.What are the main advantages and disadvantages of using a CSS framework instead of writing custom CSS from scratch?

Ans: Advantages: faster design, responsive, consistent, prebuilt components.

Disadvantages: heavy code, less customization, generic look.

24.Explain the difference between position: absolute, position: relative, and position: fixed.

Ans: 1. relative positioned relative to its normal place.

2. absolute positioned relative to nearest positioned ancestor.

3. fixed-positioned relative to the viewport, stays visible while scrolling.

25.What is the purpose of the bind () method in JavaScript?

Ans: Creates a new function with a fixed "this" value. Often used in event handlers to ensure correct object reference.

26.Describe the use of media queries in responsive web design.

Ans: Used to apply different CSS rules based on screen size or device type, making designs responsive.

27.Describe the purpose of try...catch in JavaScript. How does it improve error handling in a program?

Ans: Used to handle errors gracefully without stopping the script. Improves error handling and debugging.

28. Give an example of using for... of with an array.

Ans: `const colors = ['red', 'blue', 'green'];`

`for (const c of colors) {`

`console.log(c);`

`}`

29. What is the difference between synchronous and asynchronous JavaScript? Give one example of each.

Ans: Synchronous code runs line by line in order.

Asynchronous code (like `setTimeout` or `fetch`) runs later without blocking other operations.

30. What is the purpose of the DOM, and how does JavaScript interact with it?

Ans: The Document Object Model represents the web page structure. JavaScript can read and change elements, styles, and content dynamically.

